Abhinav Vummidichetty

APPLIED AI / ML ENGINEER

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PROFILE

AI and data engineer building multimodal search, computer vision, RAG, and agent workflows. I pair working product interfaces with reproducible evaluations and clearly stated failure modes.

TECHNICAL SKILLS

Languages and services: Python, SQL, FastAPI, React, Docker, PostgreSQL, SQLite

AI/ML: PyTorch, SigLIP 2, retrieval evaluation, PatchCore, LangGraph, Qdrant, LlamaIndex, vector search

SELECTED AI PROJECTS

MomentFlow - multimodal video moment search

- Built video upload, background indexing, visual/speech/OCR search, and timestamped playback with SigLIP 2, faster-whisper, Tesseract, SQLite FTS5, and FastAPI.
- Reached 90% Recall@1 (18/20 held-out queries) on a fixed 50-video MSR-VTT subset versus 70% for a first-frame baseline; documented 12.5% top-1 action-segment recall as a temporal limitation.

Argus - unsupervised visual inspection

- Implemented normal-only feature-bank training, coreset selection, localized heatmaps, threshold calibration, API, React console, and camera-folder gateway.
- Recorded 0.9805 image AUROC and 1.05% false-positive rate across 15 MVTec AD categories; benchmark results are not a factory deployment claim.

SeaQL - text-to-SQL agent prototype

- Built a LangGraph flow for schema retrieval, SQL generation, error-based retry, and a server-owned, short-lived, single-use review token for data-changing statements.
- Added offline approval-flow tests; remote database use still requires controlled credentials and further security hardening.

Financial Document RAG - retrieval prototype

- Combined PDF parsing, sentence-window indexing, Qdrant retrieval, Cohere reranking, and a FastAPI/React interface; identified gaps in the early six-question evaluation report.

EXPERIENCE

Data Engineering Volunteer | Beet (gobeet.app) | Jan 2026 - Present

- Built Python and SQL ETL workflows to ingest event data from 5+ sources, with schema checks and documentation for downstream data consumers.
- Designed storage schemas and partitioning strategies with product, engineering, and data science collaborators.

EDUCATION

San Jose State University - M.S. Data Analytics, expected Dec 2026 | **Andhra University** - B.Tech Computer Science, May 2024

Project metrics are scoped to the linked evaluation artifacts.